

SHOAL LAKE, MB, Canada

WMO: 711500

Lat: 50.4550N

Lon: 100.6036W

Elev: 1842

StdP: 13.74

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.4	(c) -23.4	(d) -34.6	(e) 0.8	(f) -28.4	(g) -29.2	(h) 1.1	(i) -23.4	(j) 30.0	(k) 11.2	(l) 26.7	(m) 8.6	(n) 6.4	(o) 300	(p) 0.724

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	83.2	67.8	80.1	66.1	77.2	64.2	71.6	79.2	69.0	76.4	66.6	73.9	11.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.2	114.9	75.7	66.5	104.3	73.1	64.0	95.6	70.8	36.7	79.1	34.4	75.9	32.4	74.0	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	23.0	20.5	-35.5	89.0	5.1	3.1	-39.2	91.3	-42.1	93.1	-45.0	94.8	-48.7	97.1
			-35.6	75.6	5.1	3.5	-39.3	78.2	-42.3	80.2	-45.1	82.2	-48.8	84.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	1.8	5.2	18.3	36.2	49.6	59.6	64.1	62.0	53.2	38.8	21.5	7.3	
	DBStd	24.76	14.96	14.03	14.11	11.49	8.52	6.47	5.39	6.20	7.94	8.90	12.03	13.91	
	HDD50	6872	1494	1255	983	427	112	6	0	1	56	360	855	1323	
	HDD65	11090	1959	1675	1448	864	480	184	83	132	361	811	1305	1788	
	CDD50	1382	0	0	0	13	101	293	437	374	150	14	0	0	
	CDD65	125	0	0	0	0	3	21	55	40	6	0	0	0	
	CDH74	1415	0	0	0	2	78	223	504	484	119	5	0	0	
	CDH80	272	0	0	0	0	14	37	88	116	17	0	0	0	
Wind	WSAvg	9.7	9.5	9.5	10.1	10.8	11.5	9.7	8.2	8.3	9.5	10.4	10.0	9.2	
Precipitation	PrecAvg	19.9	1.0	0.7	1.1	1.0	2.5	3.7	3.0	2.3	1.6	1.1	0.9	1.1	
	PrecMax	27.8	2.0	2.1	2.6	2.1	6.8	9.0	5.5	5.0	2.3	2.4	2.3	2.1	
	PrecMin	13.7	0.1	0.1	0.1	0.3	0.3	1.4	0.3	0.8	0.7	0.4	0.2	0.4	
	PrecStd	4.1	0.6	0.5	0.7	0.5	1.6	2.0	1.3	1.2	0.5	0.7	0.6	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	37.5	53.8	71.5	81.6	84.2	86.2	88.1	82.4	71.7	52.5	38.6	
		MCWB	33.4	35.3	45.5	53.7	59.5	67.6	72.7	69.2	62.5	55.6	45.0	35.0	
	2%	DB	31.3	33.2	44.9	64.4	75.3	79.7	82.4	83.4	77.6	63.5	45.3	33.7	
		MCWB	29.9	32.0	39.7	49.5	56.0	64.3	70.2	67.6	62.2	51.6	39.9	31.8	
	5%	DB	27.6	29.8	39.6	59.4	70.5	76.1	79.4	79.3	73.0	57.6	40.4	29.3	
		MCWB	26.7	28.8	36.3	47.2	53.9	62.1	68.4	65.6	60.1	48.4	36.8	28.1	
	10%	DB	23.7	25.7	35.7	53.9	65.8	72.6	76.4	75.8	68.0	52.5	36.5	26.1	
		MCWB	23.0	24.8	33.7	44.6	51.9	60.8	66.3	63.7	57.1	46.1	34.1	25.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.5	35.5	45.6	55.3	63.4	73.2	76.8	74.0	68.5	57.0	46.0	35.7
			MCDB	35.2	37.2	52.5	67.0	73.9	79.5	81.4	83.8	75.5	67.9	52.3	38.4
2%		WB	29.9	32.1	40.1	51.5	59.3	67.7	72.7	70.2	64.4	53.0	40.8	31.8	
		MCDB	31.2	33.2	44.6	62.0	69.8	74.3	79.0	78.8	73.0	61.7	44.3	33.6	
5%		WB	26.6	28.9	36.7	48.2	56.3	64.7	70.1	67.7	61.5	49.7	37.0	28.3	
		MCDB	27.5	29.8	39.5	58.0	66.6	72.0	77.1	75.5	70.4	55.5	40.0	29.5	
10%		WB	22.9	25.0	34.0	45.2	53.7	62.3	67.7	65.4	58.4	46.8	34.3	25.4	
		MCDB	23.5	25.9	35.8	53.2	63.5	70.2	74.4	73.3	66.1	52.6	36.6	26.2	
Mean Daily Temperature Range		5% DB	MDBR	17.1	18.2	18.6	19.9	21.6	19.3	20.1	22.5	21.6	17.8	15.2	15.5
			MCDBR	18.1	16.8	19.3	27.5	29.1	24.5	24.1	27.8	29.4	27.3	20.4	17.6
	MCWBR		17.3	15.2	14.4	15.2	13.5	12.1	13.6	14.2	15.5	16.2	15.2	16.3	
	5% WB	MCDBR	18.3	16.2	18.3	25.9	24.1	20.7	22.0	23.3	25.5	24.8	19.4	18.3	
		MCWBR	17.3	14.8	14.1	15.1	13.0	12.1	13.9	13.7	15.0	15.5	14.9	16.7	
Clear-Sky Solar Irradiance	taub	0.257	0.255	0.316	0.332	0.342	0.355	0.374	0.379	0.346	0.281	0.252	0.226		
	taud	2.252	2.294	2.228	2.382	2.415	2.410	2.371	2.345	2.414	2.573	2.483	2.374		
	Ebn at Noon	248	284	280	289	290	286	278	271	268	275	257	251		
	Edh at Noon	22	27	36	34	35	36	36	36	30	21	17	16		
All-Sky Solar Radiation	RadAvg	432	803	1245	1611	1825	1884	2006	1627	1204	712	386	305		
	RadStd	31	61	93	107	150	141	108	140	98	92	54	28		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	-0.90	N/A	N/A	N/A	N/A

Nomenclature: See separate page